

Tool #1 – Watcher Lite 1.0

Phase 1:

Problem definition

The problem I am addressing is the lack of a lightweight network monitor that can run behind a firewall to provide additional security for small subnets that are sensitive to malicious actors. A user would run this application as a secondary backup to an existing firewall. Ideally it would run on a local subnet or vlan where it can provide an additional level of security.

The problem is important because many times we rely on large firewalls to keep malicious actors out of a network but once a threat gets into a network the firewall is no longer between the two systems.

There are several tools that exist, mainly larger firewalls that consume a lot of resources. This tool has been designed to be lightweight and customizable to detect intruders via IP address and the detection of port scanning.

The gap that the tool fills is the gap between a large corporate firewall and a set of local computers (subnet or isolated vlan) that can detect IP addresses you have identified and monitor for port scans. It is not intended to replace a firewall.

System Design

The tool has two major functions. The first is monitoring for malicious IP addresses. It pulls a feed from the internet that has approximately 1,600 malicious IP addresses that have been reported. The tool currently uses IP addresses from [emergingthreats.net](https://www.emergingthreats.net). There are many sites that offer lists of bad IP addresses, and this is customizable in the code. The tool monitors traffic for these IP addresses and alerts when it finds one. The second function is that it detects port scans such as someone running nmap. It has a threshold for how many ports get scanned per second that is adjustable in the code. I chose python, scapy, and wireshark because they are lightweight, readily available on many systems, and I am familiar with them.

I used Claude AI to generate the main code for the tool. I have included it in a separate AI report file.

1. My design steps included:
 - a. Getting the main code to run correctly. This included monitoring IP traffic and detecting port scans.

- b. Adding a malicious IP address watchlist that downloads from the Internet.
The initial program had a bad_ip.txt file where you could add IP addresses manually.
- c. Adding logging to the tool.
- d. Adding an ascii interface to make monitoring more effective.

Evaluation/Testing environment

I built and tested the tool in a VMWare environment running VMWare Workstation 17.6.4. I ran the tool on my local subnet and ran various captures which produced expected results.

One limitation is that it does not detect port scans on the subnet between two machines that doesn't involve itself. This could be an enhancement to add at a later time.

The environment consisted of building an Ubuntu 24.04.4 LTS virtual machine. The steps to prepare the system include:

- `sudo apt update`
- `sudo apt install python3-pip`
- `pip install scapy requests colorama rich`
- packages may be installed using `apt install` depending on your system

The tests that I ran include:

1. Identifying IP addresses on my own subnet and verifying the tool sees them correctly.
2. Inserting a "malicious IP" address of my own to be sure the tool identifies it.
3. Performing a nmap from another computer that tests the port scan detection feature.
4. Checking the log file to be sure the alert log is

To run watcher lite:

- `sudo python3 watcherlite.py` (listen on all interfaces)
- `sudo python3 watcherlite.py -i ens33` (particular interface)
- `sudo python3 watcherlite.py -pcap file.pcap` (to scan a wireshark pcap file)

Once watcherlite runs it will display an ascii interface that shows:

- uptime/time
- top talkers on the network
- alert log
 - time

- level
- message
- threat feed: how many ip addresses it downloaded.
- the location of the log file

Screenshot of Watcher Lite 1.0 running

⚡ **Watcher Lite 1.0** | Uptime: 01:54:12 | 2026-05-23 21:33:20

Stats

Packets Analyzed	25,727
Alerts Fired	5
IPs Monitored	30
Port Scanners	1
Blocked IPs Hit	1
Scan Threshold	15 ports/10s

Top Talkers

IP Address	Packets
192.168.0.135	3512
192.168.0.1	2748
192.168.0.144	1893
192.168.0.136	1561
192.168.0.149	631
91.189.91.82	466
192.168.0.120	352

Alert Log

Time	Level	Message
19:43:17	△ ALERT	Port scan from 192.168.0.136 - 15 unique ports in 10s
19:43:03	△ ALERT	Port scan from 192.168.0.136 - 15 unique ports in 10s
19:39:15	△ ALERT	Traffic FROM malicious IP: 192.168.0.135 (192.168.0.135 → 239.255.255.250)
19:39:08	i INFO	Local IPs loaded: 3 Scan threshold: 15 ports/10s
19:39:08	i INFO	Watcher Lite started - feed: 1,651 IPs • last updated 19:39:08

⌐ Threat Feed: 1,651 IPs • last updated 19:39:08 | Log: logs/watcher_1_0.log | Press Ctrl+C to stop

Screenshot of Watcher Lite 1.0 starting and stopping

```
patrick@Ubuntu01:~/Desktop/watcherlite$ sudo python3 watcherlite10.py
[sudo] password for patrick:

⚡ Watcher Lite - starting up...

Downloading threat feeds...

Stopped. Packets analyzed: 294 | Alerts fired: 3
Full log saved to: logs/watcher_1_0.log

patrick@Ubuntu01:~/Desktop/watcherlite$
```